

All of Statistics: A Concise Course in Statistical Inference
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All errors, typographical and substantive, and other offenses, are entirely my own.

Chapter 6: Models, Statistical Inference and Learning

Question: 6.6.1

Let $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$ and let $\hat{\lambda} = n^{-1} \sum_{i=1}^n X_i$. Find the bias, se, and MSE of this estimator.

The bias, se, and MSE are given by

$$\begin{aligned}
 \text{bias}(\hat{\lambda}) &= \mathbb{E}(\hat{\lambda}) - \lambda \\
 &= \mathbb{E}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) - \lambda \\
 &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X_i) - \lambda \\
 &= \frac{1}{n} (n\lambda) - \lambda \\
 &= \boxed{0} \\
 \mathbb{V}(\hat{\lambda}) &= \mathbb{V}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\
 &= \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}(X_i) \\
 &= \frac{1}{n^2} (n\lambda) \\
 &= \frac{\lambda}{n} \\
 \text{se}(\hat{\lambda}) &= \boxed{\sqrt{\lambda/n}} \\
 \text{MSE} &= \text{bias}^2(\hat{\lambda}) + \mathbb{V}(\hat{\lambda}) \\
 &= 0 + \lambda/n \\
 &= \boxed{\lambda/n}
 \end{aligned}$$

Question: 6.6.2

Let $X_1, \dots, X_n \sim \text{Uniform}(0, \theta)$ and let $\hat{\theta} = \max\{X_1, \dots, X_n\}$. Find the bias, se, and MSE of this estimator.

Note that $\hat{\theta} \leq \alpha$ if and only if $X_i \leq \alpha$ for all $i = 1, \dots, n$. First we derive the CDF:

$$F_{\hat{\theta}}(\alpha) = \mathbb{P}(\hat{\theta} \leq \alpha) = \prod_{i=1}^n \int_0^{\alpha} \frac{1}{\theta} dx = \left(\frac{\alpha}{\theta}\right)^n$$

Differentiating gives us the PDF:

$$f_{\hat{\theta}}(\alpha) = \frac{n}{\theta^n} \alpha^{n-1}$$

Now we derive the bias:

$$\begin{aligned} \text{bias}(\hat{\theta}) &= \mathbb{E}(\hat{\theta}) - \theta \\ &= \int_0^\theta \frac{n}{\theta^n} \alpha^n d\alpha - \theta \\ &= \theta \left(\frac{n}{n+1} - 1 \right) \end{aligned}$$

Observe that

$$\lim_{n \rightarrow \infty} \text{bias}(\hat{\theta}) = \lim_{n \rightarrow \infty} \theta \left(\frac{n}{n+1} - 1 \right) = 0$$

The variance and se:

$$\begin{aligned} \mathbb{V}(\hat{\theta}) &= \int_0^\theta \frac{n}{\theta^n} \alpha^{n+1} d\alpha - \theta^2 \left(\frac{n}{n+1} \right)^2 \\ &\quad \theta^2 \left(\frac{n}{(n+2)(n+1)^2} \right) \\ \text{se}(\hat{\theta}) &= \left[\theta \left(\frac{n}{(n+2)(n+1)^2} \right)^{1/2} \right] \end{aligned}$$

Again taking the limit, we find

$$\lim_{n \rightarrow \infty} \text{se}(\hat{\theta}) = \lim_{n \rightarrow \infty} \theta \left(\frac{n}{(n+2)(n+1)^2} \right)^{1/2} = 0$$

Lastly, the MSE:

$$\begin{aligned} \text{MSE} &= \text{bias}^2(\hat{\theta}) + \mathbb{V}(\hat{\theta}) \\ &= \left(\theta \left(\frac{n}{n+1} - 1 \right) \right)^2 + \theta^2 \left(\frac{n}{(n+2)(n+1)^2} \right) \\ &= \left[\theta^2 \left(\frac{2n+2}{(n+2)(n+1)^2} \right) \right] \end{aligned}$$

with limit

$$\lim_{n \rightarrow \infty} \text{MSE} = \lim_{n \rightarrow \infty} \theta^2 \left(\frac{2n+2}{(n+2)(n+1)^2} \right) = 0$$

Question: 6.6.3

Let $X_1, \dots, X_n \sim \text{Uniform}(0, \theta)$ and let $\hat{\theta} = 2\bar{X}_n$. Find the bias, se, and MSE of this estimator.

Calculate the bias, se, and MSE as:

$$\begin{aligned} \text{bias}(\hat{\theta}) &= \mathbb{E}(\hat{\theta}) - \theta \\ &= \mathbb{E}(2\bar{X}_n) - \theta \\ &= 2\mathbb{E}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) - \theta \\ &= 2\left(\frac{\theta}{2}\right) - \theta \\ &= \boxed{0} \end{aligned}$$

$$\begin{aligned} \mathbb{V}(\hat{\theta}) &= \mathbb{V}(2\bar{X}_n) = 4\mathbb{V}(\bar{X}_n) \\ &= 4\mathbb{V}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\ &= \frac{\theta^2}{3n} \end{aligned}$$

$$\text{se}(\hat{\theta}) = \boxed{\theta/\sqrt{3n}}$$

$$\begin{aligned} \text{MSE} &= \text{bias}^2(\hat{\theta}) + \mathbb{V}(\hat{\theta}) \\ &= 0 + \theta^2/3n \\ &= \boxed{\theta^2/3n} \end{aligned}$$